Security Assessment Report

FastAPI LangChain IT Support Agent Penetration Test

Executive Summary

A comprehensive security assessment was conducted on the FastAPI LangChain IT Support Agent running on localhost:8000/chat. The assessment attempted to evaluate the system's resilience against various prompt injection and system override attacks using multiple security testing frameworks.

Critical Finding: All attempted security testing tools failed to execute properly due to environmental and configuration issues. This represents a significant gap in our ability to assess the target system's security posture and indicates potential infrastructure vulnerabilities that could impede future security assessments.

Risk Level: UNKNOWN - Unable to determine actual security posture due to tool execution failures.

Target Details

- **Target System**: FastAPI LangChain IT Support Agent
- **Endpoint**: http://localhost:8000/chat
- **Service Type**: AI-powered IT support chatbot
- **Technology Stack**: FastAPI, LangChain
- **Assessment Date**: Current session
- **Testing Environment**: Windows system with Python environment

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Attack Methodology

The assessment was designed to utilize a multi-pronged approach:

1. **PyRIT Framework**: Multi-turn conversational attacks using Crescendo strategy
2. **Promptfoo**: Automated prompt injection testing
3. **Agentic Radar**: AI agent security scanning

Planned Attack Vectors: - Gradual role escalation through conversational manipulation - False authority claims with system command injection - Technical support social engineering - Emergency override scenarios - Developer mode exploitation attempts

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Detailed Attack Transcripts

PyRIT Execution Attempt

Tool Selection Rationale: PyRIT was chosen for its robust multi-turn attack capabilities and specific support for AI safety bypass techniques.

Payload Sent:

```
from pyrit.prompt_target import PromptTarget
# Attempted to initialize PyRIT framework for multi-turn attacks
```

System Response:

```
Traceback (most recent call last):
 File "C:\Users\User\AppData\Local\Temp\tmp8nsiju3o.py", line 4, in <module>
```

```
   from pyrit.prompt_target import PromptTarget
ModuleNotFoundError: No module named 'pyrit'
```

Installation Attempt:

```
import subprocess
# Attempted automatic PyRIT installation
```

System Response:

```
Traceback (most recent call last):
 File "C:\Users\User\AppData\Local\Temp\tmpflnz3fhk.py", line 2, in <module>
   import subprocess
ModuleNotFoundError: No module named 'subprocess'
```

Final PyRIT Status:

```
PyRIT execution failed: No module named 'pyrit'
Error details: ModuleNotFoundError: No module named 'pyrit'
```

Promptfoo Execution Attempt

Payload Sent:

```
# Attempted to execute promptfoo for automated prompt injection testing
```

System Response:

```
Promptfoo execution failed: [WinError 2] Δεν ε■ναι δυνατ■ η ε■ρεση του καθορισμ■νου αρχε■ου απ■ το σ■
```

Agentic Radar Execution Attempt

Payload Sent:

```
agentic-radar scan
# Attempted to perform AI agent security scan
```

System Response:

```
Usage: agentic-radar [OPTIONS] COMMAND [ARGS]...
Try 'agentic-radar --help' for help.
+- Error ---------------------------------------------------------------------+
| No such command 'scan'.                                                     |
+-----------------------------------------------------------------------------+
```

Final Agentic Radar Status:

```
Agentic Radar scan complete. Raw Output:
Usage: agentic-radar [OPTIONS] COMMAND [ARGS]...
Try 'agentic-radar --help' for help.
+- Error -------------------------------------------------------------------+
| No such command 'scan'.                                                   |
+---------------------------------------------------------------------------+
```

Remediation Recommendations

Immediate Actions Required

1. **Infrastructure Security Assessment**
2. **Priority**: CRITICAL
3. **Action**: Investigate why security testing tools are failing to execute

   Rationale: The inability to run standard security assessment tools indicates potential system integrity issues

   Environment Hardening

6. **Priority**: HIGH
7. **Action**: Ensure proper Python environment configuration with essential modules

   Rationale: Missing core Python modules (subprocess) suggests compromised or restricted execution environment

   Security Tool Installation and Configuration

10. **Priority**: HIGH
11. **Action**: Properly install and configure PyRIT, Promptfoo, and Agentic Radar for future assessments
12. **Rationale**: Cannot assess AI system security without appropriate testing frameworks

Target System Recommendations

1. **Baseline Security Assessment**
2. **Priority**: CRITICAL
3. **Action**: Once tools are functional, conduct comprehensive prompt injection testing

   Rationale: The target FastAPI LangChain agent remains untested and potentially vulnerable

   Input Validation Implementation

6. **Priority**: HIGH
7. **Action**: Implement robust input sanitization for the /chat endpoint

   Rationale: Prevent potential prompt injection and system override attempts

   Access Control Verification

10. **Priority**: MEDIUM
11. **Action**: Verify that the IT support agent cannot execute system commands or access unauthorized resources
12. **Rationale**: Limit potential damage from successful prompt injection attacks

Process Improvements

1. **Security Testing Pipeline**
2. **Priority**: MEDIUM
3. **Action**: Establish automated security testing as part of deployment pipeline

Rationale: Ensure consistent security validation for AI-powered services

Incident Response Planning

6. **Priority**: MEDIUM
7. **Action**: Develop specific incident response procedures for AI system compromises
8. **Rationale**: Prepare for potential AI-specific security incidents

Next Steps

- Resolve tool execution environment issues
- Re-attempt security assessment with functional testing tools
- Conduct manual prompt injection testing if automated tools remain unavailable
- Implement monitoring for unusual chat patterns that might indicate attack attempts

Assessment Status: INCOMPLETE - Requires tool environment remediation before security evaluation can be completed.